

Verification o5_H_Simulator_FVM_1D_Neumann_Wells

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1 Objective

The purpose of this verification exercise is to demonstrate that the numerical simulator correctly reproduces the analytical solution for one-dimensional slightly compressible single-phase flow in a porous medium with source terms.

2 Governing Equation

For one-dimensional slightly compressible single-phase flow in a heterogeneous porous medium, conservation of mass combined with Darcy's law gives

$$\frac{\partial(\phi\rho)}{\partial t} = \frac{\partial}{\partial x} \left(\frac{\rho k}{\mu} \frac{\partial P}{\partial x} \right) + \frac{Q(x)}{V}, \quad (1)$$

where

- P is pressure [Pa],
- ρ is fluid density [kg/m³],
- ϕ is porosity [-],
- $k(x)$ is permeability [m²],
- $Q(x)$ represents source terms [kg/s],
- V is the cell volume [m³],
- μ is fluid viscosity [Pa·s].

For a slightly compressible fluid,

$$c_f = \frac{1}{\rho} \frac{\partial \rho}{\partial P}, \quad (2)$$

and assuming constant porosity and compressibility, the density is given by

$$\rho = \rho_0 e^{c_f(P-P_0)}, \quad (3)$$

where ρ_0 is the reference density at pressure P_0 . For small pressure variations, the density may be linearized as

$$\rho \approx \rho_0 [1 + c_f(P - P_0)], \quad (4)$$

which leads to the pressure diffusivity equation,

$$\phi c_f \frac{\partial P}{\partial t} = \frac{\partial}{\partial x} \left(\frac{k}{\mu} \frac{\partial P}{\partial x} \right) + \frac{Q(x)}{V}. \quad (5)$$

3 Analytical Solution

To validate the numerical simulator, a steady-state solution is used. In steady state, the flux is constant and accumulation constant. As the flux is constant everywhere in between two wells, it follows that the Darcy flux is given by,

$$q = -\frac{k}{\mu} \frac{dP}{dx} = -\frac{k}{\mu} \frac{P_2 - P_1}{x_2 - x_1}. \quad (6)$$

The pressure at any location between two wells is therefore,

$$P(x) = \frac{P_2 - P_1}{x_2 - x_1}(x - x_1) + P_1 \quad (7)$$

4 Simulation Parameters

The simulation parameters are given in Table 1 for the constant pressure well verification and Table 2 for the constant mass rate well verification.

Parameter	Symbol	Value
Number of cells	N_x	500
Domain length	L_x	10 m
Permeability	k	$1 \times 10^{-12} \text{ m}^2$
Viscosity	μ	0.1 Pa·s
Porosity	ϕ	0.2
Compressibility	c_f	$1 \times 10^{-10} \text{ Pa}^{-1}$
Reference density	ρ_0	1000 kg/m ³
Reference pressure	P_0	$1 \times 10^5 \text{ Pa}$
Well pressures	P_w	$[50, 10, 70] \times 10^6 \text{ Pa}$
Well locations	x_w	$[0, 4, 10] \text{ m}$
Wellbore indices	WI	$[1, 1, 1] \times 10^7 \text{ m}$

Table 1: Simulation parameters for constant pressure wells.

Parameter	Symbol	Value
Number of cells	N_x	500
Domain length	L_x	10 m
Cell cross-sectional area	A	1 m ²
Permeability	k	$1 \times 10^{-11} \text{ m}^2$
Viscosity	μ	0.1 Pa·s
Porosity	ϕ	0.2
Compressibility	c_f	$1 \times 10^{-10} \text{ Pa}^{-1}$
Reference density	ρ_0	1000 kg/m ³
Reference pressure	P_0	$1 \times 10^5 \text{ Pa}$
Well mass rates	Q_w	1 kg/s
Constant mass-rate well locations	x_m	0 m
Well pressures	P_w	0 Pa
Constant pressure well locations	x_w	10 m
Wellbore indices	WI	$1 \times 10^7 \text{ m}$

Table 2: Simulation parameters for constant mass rate well test.

5 Results

5.1 Constant pressure wells

Using the parameters in Table 1, the pressure solutions for both the MATLAB simulator, Figure 1, and the Python simulator, Figure 2, match well.

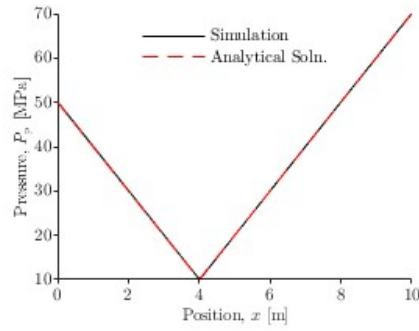


Figure 1: Comparison between analytical and numerical pressure solutions for the MATLAB simulator for constant pressure wells.

5.2 Constant mass rate wells

For the parameters, given in Table 2, the mass rate of 1 kg/s for an only slightly compressible fluid can be found as,

$$Q = -\frac{k\rho_0}{\mu} \frac{dP}{dx} A, \quad (8)$$

so that,

$$\frac{dP}{dx} = -\frac{Q\mu}{k\rho_0 A} = -\frac{1 \cdot 0.1}{1 \times 10^{-11} \cdot 1000 \cdot 1} = -1 \times 10^7 \frac{\text{Pa}}{\text{m}}, \quad (9)$$

meaning that over the 10 meters used in the simulation, a pressure drop of 100 MPa is expected. This is seen in Figure 3 and Figure 4.

Further, the mass rate in each cell should be zero, except at the inlet and outlet cells, where it should be ± 1 kg/s, as seen in Figure 5 and Figure 6.

6 Discussion

The numerical pressure distribution reproduces the analytical linear pressure solution at steady state.

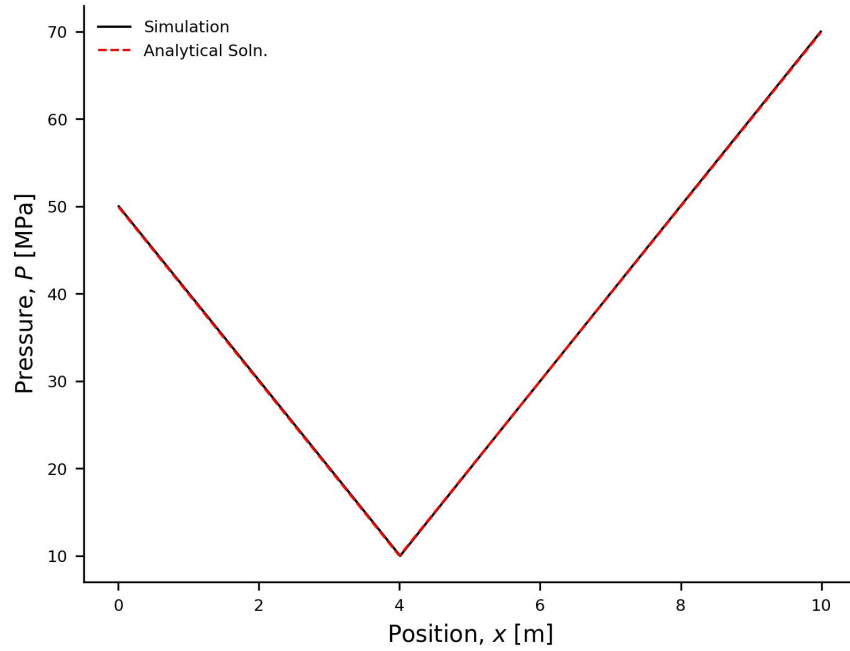


Figure 2: Comparison between analytical and numerical pressure solutions for the Python simulator for constant pressure wells.

7 Conclusion

The simulator successfully reproduces the analytical solution for one-dimensional slightly compressible flow at steady state with multiple source terms. This provides verification that the finite volume implementation is correctly solving the governing equations for this benchmark problem.

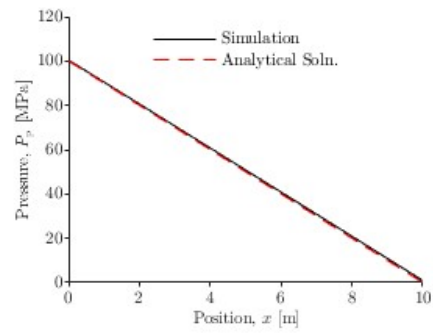


Figure 3: Steady-state pressure solution for a constant rate well in MATLAB.

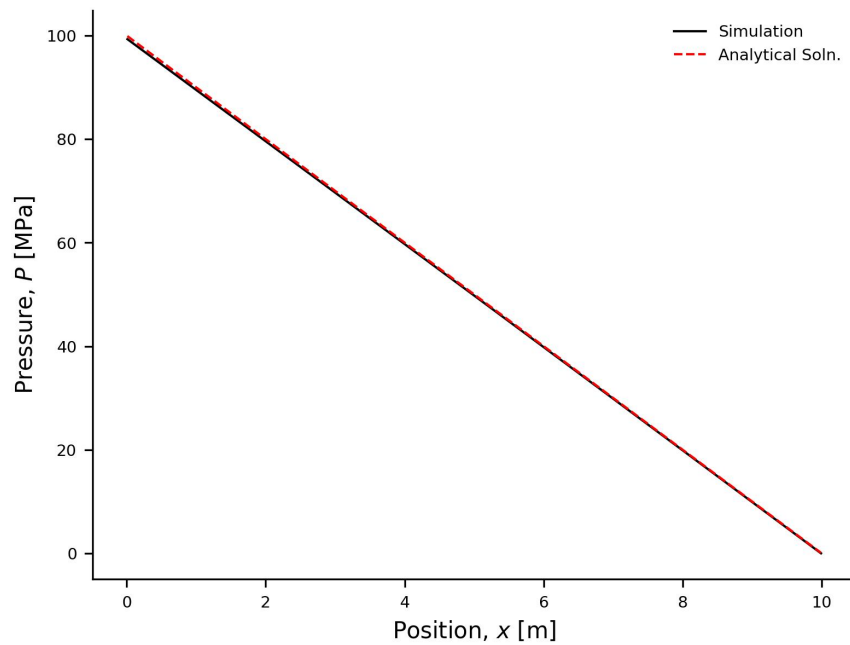


Figure 4: Steady-state pressure solution for a constant rate well in Python.

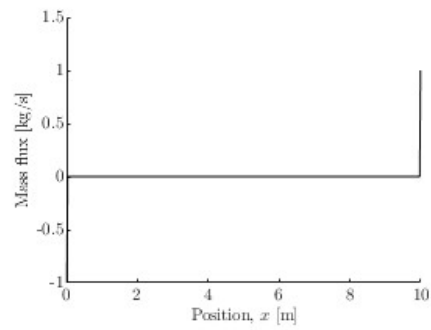


Figure 5: Steady-state mass-rate solution for a constant rate well in MATLAB.

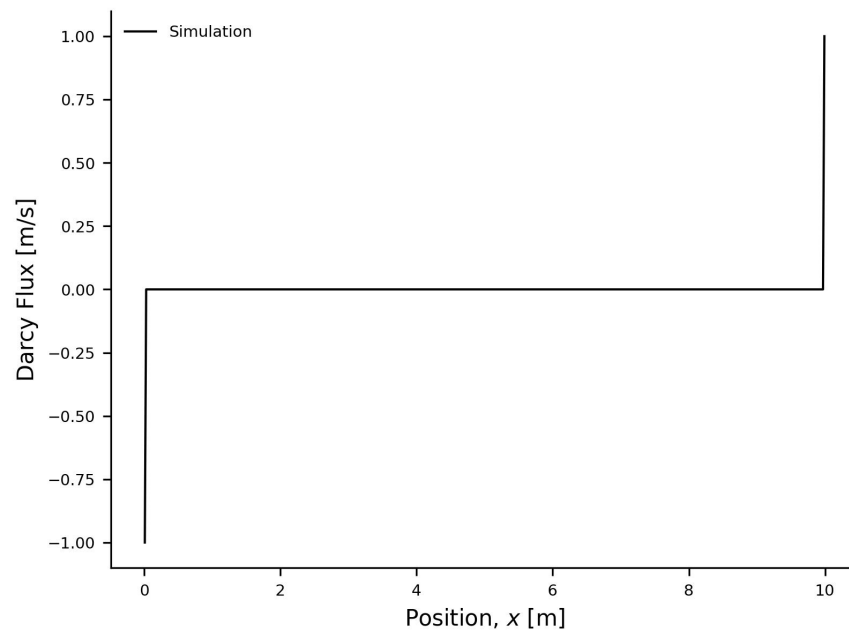


Figure 6: Steady-state mass-rate solution for a constant rate well in Python.